

Florian Grausem

PhD Candidate in Mathematics

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Research Interests

Low-Dimensional Topology, Differential/Riemannian/Hyperbolic/Symplectic Geometry, Dynamics, Algebraic Geometry, Higher Category Theory, Mathematical Physics (QFT)

Research & Projects

- Since Sept 2025 **Current Work**, *Institut de Mathématiques de Bourgogne (IMB)*
Topic: Factorization Homology for Quantum Representations of Mapping Class Groups and Quantization of Twisted Character Varieties.
In collaboration with Lukas Woike.
- Since January 2026 **Working Group Member/Speaker**, *Institut de Mathématiques de Bourgogne (IMB)*
Speaker and active participant in the research working group on **Enumerative Geometry and Mirror Symmetry**, organized by Keyao Peng.
- Since Nov 2025 **Working Group Member**, *Institut de Mathématiques de Bourgogne (IMB)*
Active participant in the algebraic geometry research working group on **Derived and Triangulated Categories**, organized by Keyao Peng.
- Since August 2025 **Current Work**, *Institut de Mathématiques de Bourgogne (IMB)*
Topic: Geometric Quantization of Character Varieties via Gauge and Chern-Simons Theory.
- 2024 – 2025 **Master's Thesis**, *Institut Montpellierain Alexander Grothendieck (IMAG)*
Topic: Big Mapping Class Groups and Realization of Isometry Groups of Infinite-type Surfaces (Grade: **17/20**)
Under the supervision of Sylvain Maillot.
- 2024 – 2025 **Academic Project**, *Institut Montpellierain Alexander Grothendieck (IMAG)*
Topic: Complex Structure of Teichmüller Moduli Spaces via the Period Coordinates of Holomorphic/Quadratic Differentials (Grade: **18/20**)
Under the supervision of Daniel Massart.

Education

- 2025–2026 **Master in Math4Phys - 2nd Year**, *University of Burgundy*, Faculty of Sciences, Dijon
Major: Mathematical Physics
- 2024–2025 **Master in Pure Mathematics - 2nd Year**, *University of Montpellier*, Faculty of Sciences
Major: Pure Mathematics
- 2023–2024 **Master in Mathematics - 1st Year**, *Sorbonne University (Pierre & Marie Curie)*, Paris
Major: Pure Mathematics and Applications.

- 2022–2023 **Bachelor in Mathematics (L3)**, *University of Lorraine*, Nancy
Major: Mathematics and Applications.
- 2021–2022 **CPGE - 2nd Year (MP)**, *Lycée François 1er*, Fontainebleau
Mathematics and Physics preparatory class.
Concurrently validated L2 Mathematics at Sorbonne University.
- 2020–2021 **CPGE - 1st Year (PCSI)**, *ECAM - Episcopal College Saint Etienne*, Strasbourg
Mathematics, Physics, Chemistry, Engineering Sciences.
- 2020 **Baccalaureate**, *French-German High School*, Saarbrücken
Majors: Mathematics & Physics.

Conferences & Seminars

- Dec 2025 **Shifted Symplectic Structures**, *Winter School*, Angers
Participant.

Experience

- Since Sept 2025 **Private Professor, Freelance**
Mathematics lessons/tutoring for High School and Undergraduate (Licence) levels.
- Mar–Jun 2025 **Research Intern**, *Institut Montpellierain Alexander Grothendieck (IMAG)*, Montpellier
Big Mapping Class Groups and Isometry Groups of Infinite-type Hyperbolic Surfaces.
- Mar 2023 **Intern**, *Institut Élie Cartan de Lorraine*, Nancy
Referencing and structuring online mathematical exercises for undergraduate students.
- Since 2021 **Tutor**, *Alpha Education*
Homework help and academic support for middle and high school students.

Skills & Languages

- Languages French (Native), English (Professional proficiency), German (Fluent)
- Computer Skills $\text{\LaTeX}$, Python

References

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Pr. Daniel MASSART
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